

1 (The Poisson distribution). Show that

$$p_\theta(x) = \frac{e^{-\theta}\theta^x}{x!}, \quad x = 0, 1, \dots$$

defines a valid probability mass function. A random variable with this p.m.f. is said to be a Poisson random variable and denoted $X \sim \text{Pois}(\theta)$.

Solution. Recall the power series

$$e^\theta = \sum_{x \geq 0} \frac{\theta^x}{x!}$$

from calculus. Notice that all terms are positive, and we can make it sum to one if we just divide the right hand side by the left hand side

$$1 = \sum_{x \geq 0} \frac{e^{-\theta}\theta^x}{x!} = \sum_{x \geq 0} p_\theta(x)$$

therefore $p_\theta = \text{Pois}(\theta)$ is a valid discrete probability distribution. $\square$

Classical statistical methods are based on the idea that the scientist has come up with the hypotheses before actually looking at the data. If the scientist looks at the data before coming up with her hypotheses, she might pick hypotheses that the data supports for no good reason at all (in certain cases this is called *p-hacking*). The typical solution to this, called *data splitting*, is to split your data into two parts, where you explore on the first one and validate your hypotheses on the second one. It turns out that this doesn't always work. An alternative to data splitting that's been developed quite recently is *data thinning*, an example of which is given in the following problem. If you want to learn more you can checkout [this paper](#).

2 (Thinning of Poisson data). Suppose that $X \sim \text{Pois}(\theta)$ and θ is unknown. Given some known ε generate $X_1 \mid X \sim \text{Bin}(X, \varepsilon)$ and $X_2 = X - X_1$. Show that $(X_1, X_2) \sim \text{Pois}(\varepsilon\theta) \otimes \text{Pois}((1 - \varepsilon)\theta)$.

Solution. Let us find the joint distribution of (X_1, X_2) . Marginalization would tell us that

$$\mathbb{P}(X_1 = x_1, X_2 = x_2) = \sum_x \mathbb{P}(X_1 = x_1, X_2 = x_2 \mid X = x) \mathbb{P}(X = x).$$

Note that since $X_2 = X - X_1$ then the only value of x that produces a non-zero value for the term inside the sum is $x = x_1 + x_2$. Therefore, we are interested in computing

$$\mathbb{P}(X_1 = x_1, X_2 = x_2 \mid X = x_1 + x_2) \mathbb{P}(X = x_1 + x_2).$$

we can approach this by expanding the first factor in the product to get

$$\begin{aligned} & \mathbb{P}(X_1 = x_1, X_2 = x_2) \\ &= \mathbb{P}(X_2 = x_2 \mid X_1 = x_1, X = x_1 + x_2) \\ & \quad \mathbb{P}(X_1 = x_1 \mid X = x_1 + x_2) \\ & \quad \mathbb{P}(X = x_1 + x_2). \end{aligned}$$

Let's evaluate each of this one by one. The first term is simply 1 because $X_2 = X - X_1$ is $x_1 + x_2 - x_2 = x_2$ when conditioning on

$$\{X_1 = x_1, X = x_1 + x_2\}.$$

The second term is

$$P(X_1 = x_1 \mid X = x_1 + x_2) = \binom{x_1 + x_2}{x_1} \varepsilon^{x_1} (1 - \varepsilon)^{x_2}$$

and the third

$$P(X = x_1 + x_2) = \frac{e^{-\theta} \theta^{x_1 + x_2}}{(x_1 + x_2)!}$$

putting it all together we obtain

$$\begin{aligned} P(X_1 = x_1, X_2 = x_2) &= \frac{e^{-\theta} \theta^{x_1 + x_2} \varepsilon^{x_1} (1 - \varepsilon)^{x_2}}{x_1! x_2!} \\ &= \left(\frac{e^{-\varepsilon \theta} \varepsilon^{x_1}}{x_1!} \right) \left(\frac{e^{-(1-\varepsilon)\theta} (1 - \varepsilon)^{x_2}}{x_2!} \right) \end{aligned}$$

Notice that the first term in this product is the p.m.f of $\text{Pois}(\varepsilon\theta)$ and the second term is the p.m.f. of $\text{Pois}((1 - \varepsilon)\theta)$. Indeed, marginalizing out each variable we obtain

$$P(X_1 = x_1) = \frac{e^{-\varepsilon \theta} \varepsilon^{x_1}}{x_1!}, \quad P(X_2 = x_2) = \frac{e^{-(1-\varepsilon)\theta} (1 - \varepsilon)^{x_2}}{x_2!}.$$

It follows that $X_1 \sim \text{Pois}(\varepsilon\theta)$, $X_2 \sim \text{Pois}((1 - \varepsilon)\theta)$, and that they are independent, which is what we wanted to show. $\square$

3 (Geometric distribution). Suppose that $X \sim \text{Geom}(p)$ and $Y \sim \text{Geom}(q)$ are independent.

- (a) Find $P(X > Y)$
- (b) If $p = q$, find $P(X \geq kY)$ for arbitrary $k \geq 1$.
- (c) If $p = q$, show that the distribution of $X \mid X + Y$ is uniform. That is,

$$P(X = k \mid X + Y = n) = \frac{1}{n - 1}, \quad k = 1, \dots, n - 1$$

Try to give a “story proof.”

Solution. Lets begin with (a). We approach this by using independence

$$P(X > Y) = \sum_{y \geq 1} P(X > y) P(Y = y)$$

Recalling the c.d.f. and p.m.f. of geometric random variables we conclude

$$\begin{aligned} P(X \geq kY) &= \sum_{y \geq 1} q(1 - p)^y (1 - q)^{y-1} \\ &= q(1 - p) \sum_{y \geq 0} [(1 - p)(1 - q)]^y \\ &= \frac{q(1 - p)}{1 - (1 - p)(1 - q)} \end{aligned}$$

which is what we wanted to find.

Onto (b), which is similar. As before we condition on the values of y and use independence

$$\begin{aligned}
P(X \geq kY) &= \sum_{y \geq 1} P(X \geq ky) P(Y = y) \\
&= \sum_{y \geq 1} (1-p)^{ky-1} (1-p)^{y-1} p \\
&= p(1-p)^{k-1} \sum_{y \geq 1} (1-p)^{y(k+1)} \\
&= \frac{p(1-p)^{k-1}}{1 - (1-p)^{k+1}}
\end{aligned}$$

Lets do (c). Note that by independence

$$P(X = k, Y = n - k) = P(X = k) P(Y = n - k) = p^2(1-p)^{n-2}$$

does not depend on k , and since $X + Y = k$ implies that

$$(X, Y) \in \{(k, n - k) : 1 \leq k \leq n - 1\}.$$

Since $X = k$ narrows it down to one of these equiprobable events, we conclude that

$$P(X = k \mid X + Y = n) = \frac{1}{n - 1}$$

as we wanted. □